

A lightweight,quality-aware IDS Framework for Highly Imbalanced IoT Environments

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Chapter 1

Introduction

1.1 Introduction

By now, the Internet of Things (IoT) has emerged as one of the most advanced evolution of modern time, connecting billions of smart gadgets to swap information, and even creating intelligent services in varied sectors like transportation, healthcare, industry, smart cities etc. But the sprawling IoT networks now pose a great challenge to security with a wide range of devices with limited computational capabilities, connectivity and they are becoming smarter in perpetrating cyber-attacks.

Intrusion Detection Systems (IDSs) are key protectors of IoT environment which try to identify any malicious activities and block access to unauthorized people. Machine learning and deep learning methods have recently emerged as useful tools in enhancing the IDS performance. Though the performance of these IDS has often been undermined by the class imbalance problem in IoT datasets, This imbalance can lead to biased models that fail to accurately detect minority attack classes.

To address this challenge, researchers have increasingly used generative artificial intelligence techniques for generating synthetic data to balance datasets. but , most existing approaches focus primarily on increasing the quantity of generated data without adequately assessing its quality and reliability. Furthermore, many proposed solutions require considerable computational resources making them unsuitable for deployment in resource-constrained IoT devices. Therefore, this study proposes a lightweight, quality-aware AI framework designed to improve intrusion detection performance while ensuring efficient operation in real-world IoT environments.

1.2 Research Problem Statement

The rapid spread of IoT technology has created significant security challenges, particularly in the development and deployment of effective intrusion detection systems. One of these ongoing problems and obstacles is the significant imbalance between classes in network traffic, leading to the overlooking of lesser-represented attack classes by traditional machine learning classifiers. To address this, generative artificial intelligence was used to generate artificial samples for lower

classes. However, the primary focus when creating this new data is mostly on the quantity of data generated rather than its quality. That is, it depends on the volume of data generated instead of applying precise metrics to measure reliability and conformity. Also, most of these solutions are designed to be suitable for high-performance environments without considering the limitations of resources such as processing power and energy consumption in actual IoT devices. Thus, we discover two research gaps in this field: the lack of quantitative metrics to evaluate the quality of the generated data, and the scarcity of lightweight models that are suitable for the resource-limited IoT environment. As a result, current intrusion detection systems suffer from substandard accuracy and difficulty in deployment in real IoT devices, leaving the IoT environment vulnerable to cyberattacks and low operational efficiency.

1.3 Purpose Statement

The purpose of this quantitative experimental study is to develop and evaluate a lightweight, quality-aware AI framework to address the imbalance problem in IoT intrusion detection systems. Specifically, this study employs a set of quantitative metrics to accurately assess the consistency and reliability of generated artificial intelligence data. It also aims to verify the model's accuracy in detecting attacks and its computational efficiency with limited resources in a real-world IoT environment.

1.4 Main Research Question

How can a lightweight, quality-aware AI framework improve the accuracy of attack detection and reduce the false positive rate in IoT intrusion detection systems on real-world, resource-limited IoT devices?

1.5 Sub-questions

1. How accurate will the proposed lightweight, data-quality-aware, generative AI-based framework be in detecting attacks and achieving a false positive rate when applied to an unbalanced IoT traffic dataset?

2. How effective is the assessment of the quality of data generated by using quantitative metrics in building an effective and trustworthy intrusion detection model?

3. How does the proposed generative framework compare with other traditional techniques for addressing class imbalances, such as smote?

4. When deploying the proposed generative framework on (resource-limited) real Internet of Things devices, what is the level of computing resource consumption?

1.6 General Objective

To develop and evaluate a lightweight, quality-aware AI framework to address the imbalance problem in IoT intrusion detection systems. Specifically, this study employs a set of quantitative metrics to accurately assess the consistency and reliability of generated artificial intelligence data. It also aims to verify the model's accuracy in detecting attacks and its computational efficiency with limited resources in a real-world IoT environment.

1.7 Specific Objectives:

1. Design and development of a generative AI framework to address class imbalances in IoT intrusion detection datasets that is lightweight and data quality-focused.
2. Evaluate the impact of generated synthetic data on the performance of an intrusion detection model, specifically detection accuracy and false positive rate.
3. Applying a set of precise quantitative metrics to assess the quality of artificially generated data to ensure the accuracy of the prediction is not truly necessary.
4. Verifying the computing efficiency of the proposed framework used in a resource-limited IoT environment by measuring resource consumption such as processor, power, and memory.
5. Comparing the proposed generative AI framework with traditional approaches that address the problem of class imbalance, such as SMOTE.

1.8 Significance of the Study:

This study is important in three dimensions:

- **Theoretical:** It advances IoT security research by introducinThe study contributes to the development of research related to IoT security by developing an artificial intelligence framework for intrusion detection systems to address the imbalance of classes in IoT datasets by shifting the focus to data quality rather than the quantity of artificially generated data.
- **Practical:** Providing a lightweight, quality-aware Intrusion-Detection Framework for Highly Imbalanced and resource-limited IoT Environments.
- **Methodological:** Integrating quality assessment criteria for the generated data with computational efficiency metrics of the proposed model makes it a benchmark model for future research to build upon.

1.9 Scope & Delimitations

Scope of the Study:

The scope of this study is limited to building and evaluating a lightweight, data-aware, generative AI framework for IoT environments. The goal of this framework is to address class imbalances in IoT intrusion detection systems.
The study focuses on datasets with significant imbalances between natural and attack classes. It also includes a set of quantitative evaluation criteria to measure the quality of the generated synthetic data, along with traditional methods for measuring detection accuracy , false positive rates and F1 score. Furthermore, the study assesses computational efficiency using a set of metrics to measure resource consumption in resource-constrained IoT devices. This framework will be compared with traditional methods for generating synthetic data to address class imbalances such as SMOTE.

Delimitations of the Study:

The study is limited by a number of constraints, including: the proposed framework is not deployed on real-time IoT devices, the study uses a specific standard dataset and therefore cannot be generalized to all IoT devices. The study is limited only to building a framework that focuses primarily on the quality of synthetic data generated in the unbalanced dataset and also trying to use a lightweight model that is suitable for a resource-limited IoT environment.

1.10 Terminology Definitions:

1. **Generative Artificial Intelligence (GenAI):** is the name given to a subset of AI machine learning technologies that have recently developed the ability to rapidly create content in response to text prompts, which can range from short and simple to very long and complex.

2. **Class Imbalance:** Refers to the irregular data distribution between classes, where one class contains a disproportionate number of instances compared to the other. This can lead to difficulties in training models and lower accuracy in object detection[1].

3. **Internet of Things (IoT):** Refers to a network of physical devices, vehicles, appliances, and other physical objects that are embedded with sensors, software, and network connectivity, allowing them to collect and share data[2].

4. **Intrusion Detection System (IDS):** is a network security tool that monitors network traffic and devices for known malicious activity, suspicious activity or security policy violations[3].

5. **Synthetic Data:** Artificial data designed to mimic real-world data. It's generated through statistical methods or by using artificial intelligence (AI) techniques like deep learning and generative AI[4].

6. **Lightweight Model:** Refers to models designed to be compact, efficient and less demanding in terms of computational resources.
   They are optimized to run on devices with limited capabilities, such as smartphones, IoT devices, or hardware with memory and processing constraints[5].

7. **SMOTE (Synthetic Minority Over-sampling Technique):**is a method for oversampling samples when dealing with imbalanced classification categories within a dataset[6].

8. **Detection Accuracy:** Refers to the proportion of correct predictions made by a model out of all predictions. It is a straightforward evaluation metric that indicates how accurately a model is able to make predictions[7].

9. **False Positive Rate (FPR):** is the probability of falsely rejecting the null hypothesis for a particular test. The false positive rate is calculated as the ratio between the number of negative events wrongly categorized as positive (false positives) and the total number of actual negative events (regardless of classification)[8].

1.11 Limitations of the Study

1. **Dataset Generalizability:** In this study, we use a standard, general-purpose dataset used to train and evaluate the proposed framework. Although this dataset is widely known in research circles, it does not reflect the full picture of the changing nature of IoT traffic in the real environment across different sectors. Therefore, the model's performance may differ when applied to an environment with traffic that differs from that used in the study.

2. **Handling Zero-Day Attacks:** The proposed framework is used to generate synthetic samples based on patterns learned from known attacks contained within the training data, and therefore the model may face limitations in detecting or generating samples for zero-day attacks.

3. **Complexity of Real-World Environments:** The proposed framework is used to generate synthetic samples based on patterns learned from known attacks contained within the training data, and therefore the model may face limitations in detecting or generating samples for zero-day attacks.

1.12 Research Hypotheses:

1. **H1:** A proposed lightweight AI framework that focuses on the quality of generated data will significantly improve the accuracy of attack detection and reduce the false positive rate when compared to traditional models that address class imbalances, such as SMOTE.

2. **H2:** There is a statistically significant positive correlation between the quality of generated synthetic data and the performance of intrusion detection model.

3. **H3:** The proposed framework will achieve greater computational efficiency in terms of resource consumption when compared to other deep learning-based models when deployed on resource-limited IoT devices.

4. **H4:** The proposed data-quality framework outperforms traditional generative models that focus on the quantity of data rather than its quality when dealing with underrepresented attack classes in the IoT environment.

1.13 Brief Methodology Overview

This study follows a quantitative experimental approach to validate the proposed lightweight and quality-aware generative AI framework. The stages used in the research methodology:

1. **Data Preparation:** Selecting dataset and checking for infinite and missing values and filling them with mean values. The data are then normalized , and the data set is divided in to training and testing datasets.

2. **GAN Architecture Design:** We adopt a Conditional Generative Adversarial Network (cGAN) framework to generate synthetic samples conditioned on the class label, enabling targeted minority class data augmentation.

    - **Generator:** A deep neural network that receives noise vectors concatenated with the minority class label as input and outputs synthetic feature vectors resembling real intrusion samples.
    - **Discriminator:** A binary classifier that receives both real and generated samples, conditioned on the class label, and learns to distinguish authentic from synthetic data.
    - Using quality assessment criteria for generated synthetic data to ensure high-quality synthetic data.

3. **Experimentation:** Training and testing intrusion detection systems using the augmented dataset, and comparing the proposed model with traditional synthetic data generation methods such as Smote.

4. **Evaluation:** The final step is to use a set of evaluation criteria to assess the proposed model, which will be evaluated from two aspects: detection effectiveness and computing resource efficiency.

Chapter 2

Literature Review

2.1 Introduction

Due to the widespread adoption of Internet of Things (IoT) devices, many have become vulnerable to cyberattacks. Therefore, it is essential to use technologies to protect the IoT environment. One such technology is intrusion detection systems, which act as a defensive layer, detecting suspicious and malicious activities and any unauthorized access. However, when we transition from traditional networks to the IoT, we face a number of additional challenges. The IoT environment is characterized by limited resources, and the nature of attacks is more complex.

Machine learning and deep learning techniques have been used to improve intrusion detection systems by detecting anomalous and previously unknown activities in network traffic. However, despite this progress, these models are significantly affected by the nature of the data in IoT traffic, particularly the problem of class imbalance. The proportion of attack samples is much lower than that of normal samples, leading to a bias in the models towards normal categories and resulting in the failure of these systems to detect attacks. While generative AI techniques have been developed to generate artificial samples of under-represent categories, most of these models focus on the quantity of generated artificial data rather than its quality through evaluation. Furthermore, some studies have not focused on building lightweight models suitable for resource-constrained IoT environments, hindering their deployment in such environments.

This chapter will survey and review a selection of previous studies related to IoT security, addressing class imbalance in datasets, and the generative AI techniques used in intrusion detection systems. A critical analysis of each paper will be conducted, focusing on the strengths and weaknesses of each study and its methodology.

2.2 Literature Review

Analysis of previous studies related to generative artificial intelligence for intrusion detection systems in an IoT environment:

Table 2.1: General details of selected studies (2023–2025)

ID	Study	Year	GAN Type	Datasets	Classifiers
S1	GAN&CNN [9]	2025	GAN + CNN	UNSW-NB15, CICIDS2018	CNN
S2	FL-BiLSTM [10]	2025	WGAN	CICIoT2023, BoT-IoT	LSTM+BiLSTM
S3	CTGAN-ENN [11]	2025	CTGAN	CIC-IDS2018	RF+XGBoost
S4	Minority-GAN[12]	2025	CGAN	CICIDS2017	RF+MLP
S5	CWGAN-GP [13]	2025	CWGAN-GP	NSL-KDD	XGBoost+LR
S6	FIGS [14]	2025	CGAN+SMOTE	CICIoMT2024	Ensemble
S7	SNNL-GAN [15]	2025	WGAN+SNNL	IDS2017	KNN
S8	Create a Realistic IoT Dataset Using CGAN [16]	2024	CGAN	Bot-IoT, Joint IoT Dataset	(no specific classifier)
S9	Comparative Analysis of TDCGAN Model [17]	2024	TDCGAN	CIC-IDS2017,CSE-CIC-IDS2018,KDD-Cup 99,Bot-IoT	DT,RF,MLP,Naive Bayes
S10	Tree-Based [18]	2023	No GAN	CICIDS2018	Ensemble

Table 2.2: Comparative Analysis of Recent Studies on Class Imbalance Mitigation in IoT IDS

ID	Abstract	Methodology	Results	Strengths	Weaknesses and Gaps
S1	Lightweight hybrid model combining GAN and CNN for IoT intrusion detection.	• GAN for minority classes. • Lightweight CNN with HSwish and CSP. • Datasets: UNSW-NB15, CICIDS2018.	• 99.64% accuracy. • 99.65% accuracy. • Size: 21–32 KB.	• Very lightweight. • High accuracy.	• No quality evaluation. • No real device testing. • No SMOTE comparison.
S2	Lightweight decentralized IDS using FL, WGAN, and BiLSTM.	• K-means clustering. • Autoencoder reduction. • Lightweight BiLSTM.	• 99.5% accuracy. • Model size 60 KB. • Better DDoS detection.	• Efficient. • Handles heterogeneity.	• Poor WGAN F1. • No quality metrics. • No real deployment.
S3	Hybrid CTGAN + ENN + XAI framework.	• CTGAN generation. • ENN cleaning. • SHAP, LIME.	• 99.99% accuracy. • Better than SMOTE. • Optimal ratio 60–80%.	• Strong comparison. • Explainability.	• Not lightweight. • No real device test.
S4	GAN for rare attack classes (U2R, R2L).	• Conditional GAN. • Merged synthetic data.	• U2R Recall improved. • R2L Recall improved.	• Focus on rare classes. • Better than SMOTE.	• No quantitative quality test. • No lightweight design.
S5	CWGAN-GP for mode collapse and stability.	• Wasserstein distance. • Gradient penalty. • Conditional generation.	• Better stability. • Better multiclass results.	• Solves training instability.	• No direct quality metric. • No lightweight model.

ID	Abstract	Methodology	Results	Strengths	Weaknesses and Gaps
S6	FIGS framework integrating GAN + SMOTE + feature importance.	• GIR metric. • Feature importance. • FIGAN + FISMOTE.	• 17% improvement. • Bot F1 to 0.99.	• Novel method. • Strong rare class results.	• No real device test. • No KL metric.
S7	GAN quality improved using SNNL.	• SNNL regularization. • UMAP visualization.	• Better minority F1. • +25% training cost.	• Novel alignment method. • Visual evidence.	• Heavy GPU models. • No real device testing.
S8	CGAN-based tool for realistic IoT dataset generation.	• Conditional GAN used to generate synthetic IoT traffic. • Applied on Bot-IoT and custom Joint Dataset. • Includes Exploratory Data Analysis + feature ranking.	• Reduced imbalance in IoT datasets. • Generated high-quality synthetic traffic. • Improved dataset diversity for IDS training.	• Provides a novel dataset generator tool. • Focus on realistic IoT attack scenarios. • Enhances research resources for IDS..	• No deployment on real IoT devices. • Focused on dataset generation, not classifier performance. • Computational efficiency not evaluated.
S9	TDCGAN model for data balancing and IDS performance.	• Triple Discriminator Conditional GAN (TDCGAN). • Tested on CIC-IDS2017, CSE-CIC-IDS2018, KDD-Cup 99, Bot-IoT. • Compared classifiers: DT, RF, MLP, Naive Bayes.	• Significant improvement in accuracy and F1-score after balancing. • Better minority class detection. • Up to 95% accuracy on UGR'16 dataset.	• Strong comparative analysis across multiple datasets. • Novel ensemble-like GAN design (three discriminators + election layer). • Demonstrated clear benefit of balancing.	• High computational cost.. • Complex architecture, not lightweight. • No real-time IoT deployment tests.
S10	Traditional IDS using tree models + SMOTE.	• SMOTE balancing. • Information Gain. • RF, DT, XGBoost.	• 98.36% accuracy. • Improved performance.	• Strong baseline. • Good methodology.	• No GAN. • No quality evaluation. • No lightweight focus.

The current focus is on identifying trends in studies related to the problem of class imbalance in intrusion detection systems within the Internet of Things (IoT). The analysis of these studies focused on the methodologies used, particularly the application of generative adversarial networks (GANs), as well as traditional sampling methods. The efficiency and effectiveness of these models were evaluated when handling unbalanced IoT network traffic. Some studies employed lightweight

models, others used hybrid approaches, and still others utilized improved GAN models in terms of stability.

2.2.1 Firstly, lightweight methodologies with high computational efficiency:

Some studies have prioritized lightweight methodologies to suit the resource-limited nature of the IoT environment. S1 presented a hybrid model combining generative adversarial networks and convolutional neural networks with the HSwish activation function and CSP architecture. This resulted in a model with a small size of 21–32 KB, while maintaining a high accuracy of 99.64% on the UNSW-NB15 dataset.

Another study, S2, presented a federated learning model that integrates Wasserstein GAN (WGAN) with BiLSTM to address device variability in the IoT environment. Although the WGAN-AE component was described as small (60 KB), the study revealed a clear performance trade-off, with the WGAN achieving a low F1 score of 0.3957, indicating that excessive size reduction may negatively impact generation quality.

S8 (CGAN-IDS, 2024) contributes by generating balanced synthetic IoT datasets using Conditional GAN (CGAN). However, it was not tested on resource-constrained IoT devices, so it cannot be considered a fully lightweight methodology.

2.2.2 Hybrid and Feature-Driven Generative Models:

To obtain useful synthetic data, some research has focused on developing hybrid frameworks that combine feature selection and interpretability. Study S3 combined Conditional Tabular GAN (CTGAN) and Edited Nearest Neighbors (ENN) with interpretable artificial intelligence (XAI) tools such as SHAP and LIME. This study provided a comprehensive comparison with SMOTE and determined the optimal synthetic data ratio to be between 60% and 80%. However, this study did not address the need for a small model size to suit resource-limited IoT environments, making this model impractical for deployment in real-world IoT devices.

Another model, Study S6, presented the FIGS framework, which integrates feature significance analysis during generation. By generating data only for significant features using a combination of GAN and SMOTE, the framework achieved exceptional improvements in detecting rare classes, increasing the F1 score for the Bot class from 0.54 to 0.99. The study relied on GIR (Generalized Inequality Ratio) metrics instead of standard quantitative quality metrics like KL Divergence to assess data quality.

Study S7 focused on improving intraclass coherence by integrating Soft Nearest Neighbor Loss (SNNL) into WGAN evaluator training. UMAP visualizations showed improved alignment of mi-

nor classes like U2R, but this model resulted in a 25% increase in computing costs and requires dedicated GPUs, making it impractical for traditional IoT devices.

S9 (TDCGAN-IDS, 2024) fits in this category because it employs a Triple Discriminator Conditional GAN with an election layer and compares the performance of multiple classifiers (DT, RF, MLP, NB). This enhances understanding of how balancing impacts classifier performance.

2.2.3 Stability and Handling Rare Classes:

Some studies have focused on addressing the problem of rare attack types. Study S4 used Conditional Generative Adversarial Networks (CGANs) to generate samples of highly rare classes such as U2R and R2L. The results showed a relative improvement in recall, with the U2R value increasing from 0.42 to 0.78. However, the data quality assessment relied on visual t-SNE plots without using quantitative measures of statistical accuracy.

Study S5 applied the CWGAN-GP model to address training instability and mode collapse. The result was that it improved the stability of the training process, but the study did not evaluate the quality of the generated data, nor did it assess the model's computational efficiency.

S8 addresses this challenge by generating realistic synthetic data that reduces imbalance, although it did not evaluate classifier performance directly.

S9 demonstrated significant improvement in detecting minority classes after balancing with TDCGAN, proving its effectiveness in stabilizing detection across multiple datasets.

2.2.4 Baseline Comparisons

Some S10 studies presented a baseline using Random Forest and XGBoost algorithms with SMOTE technology and feature selection using Information Gain. The result was an accuracy of up to 98.36

2.2.5 Conclusion of the most important research gaps:

Through the comprehensive analysis of scientific papers, a set of research gaps were discovered, including:

1. **The gap of lack of attention to the quality of generated synthetic data:** Most studies did not pay attention to the quality of generated synthetic data and did not conduct an evaluation of data quality through quantitative statistical criteria such as Frechet Inception Distance (FID), KL Divergence, or Wasserstein Distance to ensure that the synthetic data is identical to the real data, but rather they were satisfied with relying on the final classifier accuracy as a criterion for improvement.

2. **The gap in deployable lightweight models in the real environment:** Many studies have not paid attention to developing small-scale models to suit deployment in resource-limited IoT devices. Also, even studies that have resulted in small-scale models have not applied these models to real devices in the real environment to measure the effectiveness of these models.

3. **Feature integration gap:** The problem of class imbalance usually exists between data, but this problem is usually increased by the presence of unimportant features or irrelevant characteristics. There has been only one study that relied on the importance of features in the generation process S6, and therefore here is a research opportunity to enhance generative models by selecting important features not only to improve the classifier but also to improve the generation process.

2.3 Conclusion of the review of previous studies:

Through the comprehensive analysis of previous studies, it was shown that generative adversarial networks are significantly superior to traditional methods for addressing the problem of class imbalance by generating synthetic data. Through the studies, it was shown that there is no integrated framework that combines the use of quantitative criteria to measure the quality of the generated synthetic data while at the same time trying to maintain a lightweight model with a small size, and this is where the role of this study comes in.

Chapter 3

Research Methodology

3.1 Introduction

This chapter transitions from the theoretical framework presented in previous chapters to a structured, experimental process, serving as the research blueprint. The primary objctive is to present the structured methodolgy for the proposed lightweight AI-based framework focused on the quality of data generated for intrusion detection systems in an IoT environment. This chapter will outline the research design, the nature of the study, data collection methods, the experimental environment, and the comprehensive metrics used to verify the research hypotheses.

3.2 Research Design and Study Nature:

To address the research questions and hypotheses efficiently and effectively, this study will adopt a comparative evaluation design. The performance evaluation of the proposed framework will be compared with basic traditional methods (baselines).

1. **Research Nature "Second Track: Algorithm/System Evaluation " :** This study falls under the category of " Algorithm/System Evaluation " which focusing on the numerical performance of a generative AI-based intrusion detection system model. The research focuses on a standard dataset and benchmarks.

2. **Unit of Analysis:** The basic unit of analysis is the network traffic state (individual packets or flow record).

3. **Variables:**

   (a) **Independent:** A synthtic data generation enhancement technique that is aware of the quality of the generated data compared to SMOTE and other GAN techniques, with a lightweight CNN classifier architecture.

   (b) **Dependent:**
   Detection Accuracy: Accuracy score, F1 score, False positive rate.
   Generated Data Quality metrics: KL Divergence, TSTR.

Computational Efficiency metrics: Measured power consumption, model size, inference time.

3.3 Dataset:

We will use a standard, general dataset containing unbalanced IoT traffic data:

1. **UNSW-NB15**
   This dataset contains a mix of normal traffic and modern attacks.

   - **Reason for choosing this dataset:** It has proven suitable for lightweight models and contains nine types of attacks.

2. **CICIDS2018**
   We will use it to verify generalizability and it includes HTTPS traffic and modern attacks such as botnets.

Data Splitting:

- 70% Training
- 15% Validation
- 15% Testing
- **Decision Rule:** Rare attack class ratios must be maintained across all divisions to preserve the "unbalanced" nature of the problem.

3.4 Proposed Framework Architecture

The Framework consists of two main units:

3.4.1 Data Quality-Based Genrative Adversarial Network:

The generative adversarial network consists of two networks: a generator and a discrminator. Quality criteria are added during the generation process, allowing the genrated data to be filtered based on these criteria. Only high-quality data that matches real-world samples is retained.

3.4.2 Lightweight CNN Classifier:

This model is designed to work in resource-constrained IoT environments. We will minimize the computational load by:

1. Cross Stage Partial architecture will be used, which divides the features in the base layer into two parts and processes them across separate, dense blocks. This helps reduce computations by minimizing gradient duplication.

2. Using the HSwish activation function instead of Relu

3. Global Average Pooling (GAP): Instead of using fully connected layers with large parameters, GAP is used in the end layers to flatten the feature maps, significantly reducing memory consumption.

4. Depthwise Separable Convolutions Traditional convolutions will be replaced by this type to reduce the number of computations , which helps to reduce the model size to less than 100 KB.

3.5 Experimental Environment

This section describes the technical environment used in developing and implementing this framework to ensure the reproducibility of results and the fairness of comparative analysis:

1. **Hardware:**

   - During Development: We will use a device with a GPU such as an AMD Radeon HD to train the GAN and CNN models. This environment provides the necessary computing power for generation.
   - During Deployment: The trained model will be deployed on a Raspberry Pi 4 Model B (4GB RAM and an ARM Cortex-A72 processor). This device will be used to simulate and verify the typical computing power of IoT devices, ensuring the model is lightweight.

2. **Software:**

   - Operating System: with development we use(Windows 10) , with deployment we use(Raspberry Pi OS).
   - Software Frameworks: Python 3.12.7 , libraries (TensorFlow/Keras, Scikit-learn, and Pandas/NumPy).

3. **Reproducibility Controls:**The seed value is a constant value in all libraries to ensure that the results are consistent and reproducible across multiple runs (e.g., seed=42).

3.6 Experimental Protocol, Baselines, and Operational Definitions

The experimental protocol, methodology, and comparative evaluation framework were designed systematically. To ensure verifiability and reproducibility, and in accordance with the quantitative study guidelines, a set of operational definitions and decision-making rules were clearly defined:

Experimental Set (Baseline): The proposed framework will be evaluated with three different baseline models.

1. SMOTE(Sampling of Underrepresented Categories) using a traditional classifier as the standard reference for addressing the class imbalance problem.
2. Using a traditional competitive generative network (GAN) without any optimization or consideration of the quality of the generated synthetic data, to observe the impact of model improvement when evaluating the quality of the generated data
3. The proposed framework will be used in the S1 study as a modern SOTA reference for evaluating the lightweight model and its performance efficiency.

Operational Definitions and Decision-Making Rules (Processing Strategy):

1. **Step 1:** Initial Data Processing: A preliminary step before training any model is data cleaning and processing. The following processes will be performed on the data before training:
    - **Rule 1: Handling Missing Values:** If the percentage of rows containing missing values is small (less than 5%), these rows will be deleted. If the percentage exceeds this, the missing values will be replaced with the column mean for numeric data or the column model for textual data.
    - **Rule 2: Converting Textual Columns to Numeric Data:** One-hot encoding will be used to convert certain properties, such as class properties like protocol type, to match the neural network inputs. Label encoding will also be used to convert other properties, such as attack names.
    - **Rule 3: Data Normalization:** The Min-Max function will be used to ensure all property values range between 0 and 1 to guarantee efficient convergence of the CNN model.
    - **Rule 4: Class Encoding:**The data will be encoded (normal data 0, attack data 1).
2. **Step 2:** Generating Synthetic Data.
    - **Rule 5:** After processing and cleaning the data, the generation unit is trained on the least representative categories to attempt to generate synthetic data from these categories.

- **Rule 6:** After data generation, the generated data is passed through data quality metrics. If the generated data meets a certain threshold, it is accepted; otherwise, it is rejected.

3. **Step 3:** Training:

   - **Rule 7: Model Training:** The lightweight CNN model is trained on the real dataset and the accepted synthetic data.
     The validation set contains only real data to monitor for the possibility of overfitting.

4. **Step 4:** Model Deployment and Inference:

   - **Rule 8:**The model is deployed on a real device, as previously described (Raspberry Pi).
   - **Rule 9: Inference Recording:** When the test dataset, which contains only real data, is categorized, resource usage indicators such as CPU, memory, and power consumption are recorded to later support our claim of model efficiency.

3.7 Evaluation Criteria and Research Plan:

We will use a variety of evaluation criteria to assess the research objectives (a lightweight model, and an awareness of the quality of the generated synthetic data). Each evaluation criterion will be linked to the appropriate research question.

Attack Detection Performance Criteria (IDS Efficiency):

We will use a set of criteria to evaluate the accuracy and capability of the CNN classification algorithm for classifying and detecting attacks.

1. (Accuracy score, Precision, Recall, F1 score) These are the criteria for evaluating classification algorithms.

2. False Positive Rate: The percentage of legitimate traffic that is incorrectly classified as attack traffic.

Evaluation Criteria for the Quality of the Generated Synthetic Data:

To address the lack of attention given to evaluating the quality of the generated synthetic data, the following quantitative measures will be used.

1. Training on Synthetic Data and Testing on Real Data (TSTR): The model is trained on synthetic data, while only real data is used during the testing phase. If the accuracy is high during the testing phase, this indicates that the generator has learned the property distribution correctly.

2. KL Divergence: This metric is used to measure the degree of divergence between the probability distribution of generated synthetic data and real data.

3. t-SNE Algorithm: This algorithm is used to reduce the dimensions of data representation in a two-dimensional space in order to see the degree of overlap between generated synthetic data and real data.

Computing Efficiency Assessment Criteria:

To verify the feasibility of deploying the model or framework on real-world, resource-limited IoT devices, a set of hardware-related inferences will be measured on a Raspberry Pi.

1. MODEL SIZE: Calculate the storage space required for the model weights (we want a model smaller than 100 KB).

2. INFERENCE TIME: How long is required to process a single network flow?

3. PROCESSOR USAGE: What percentage of the processor is used during the inference process?

4. MEMORY USAGE: How much memory is used during operation?

5. POWER CONSUMPTION: How much power is consumed especially for battery-powered devices?

3.8 Ethical Considerations:

Although algorithms are used in this study and do not rely on human input, several ethical considerations related to data use must be taken into account.

1. DATA PRIVACY: The study uses publicly available, anonymized datasets (UNSW-NB15 and CICIDS2018) that comply with their respective license agreements.

3.9 Potential threats That May Affect the Validity of the Results

This section outlines potential constraints that may affect the validity of the experimental results.

1. Internal Validity: To minimize the random nature of training adversarial generative adversarial networks (GANs), the experiment will be repeated multiple times to ensure the validity of the results . The arithmetic mean and standard deviation of the results will be reported to ensure that the conclusion is based on evidence.

2. External Validity (Generalizability): The results are specific to the UNSW-NB15 and CICIDS2018 datasets. Although these datasets represent realistic scenarios, the model's performance may differ in real-world production IoT environments with significantly different traffic patterns or novel attacks not present in the training dataset.

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